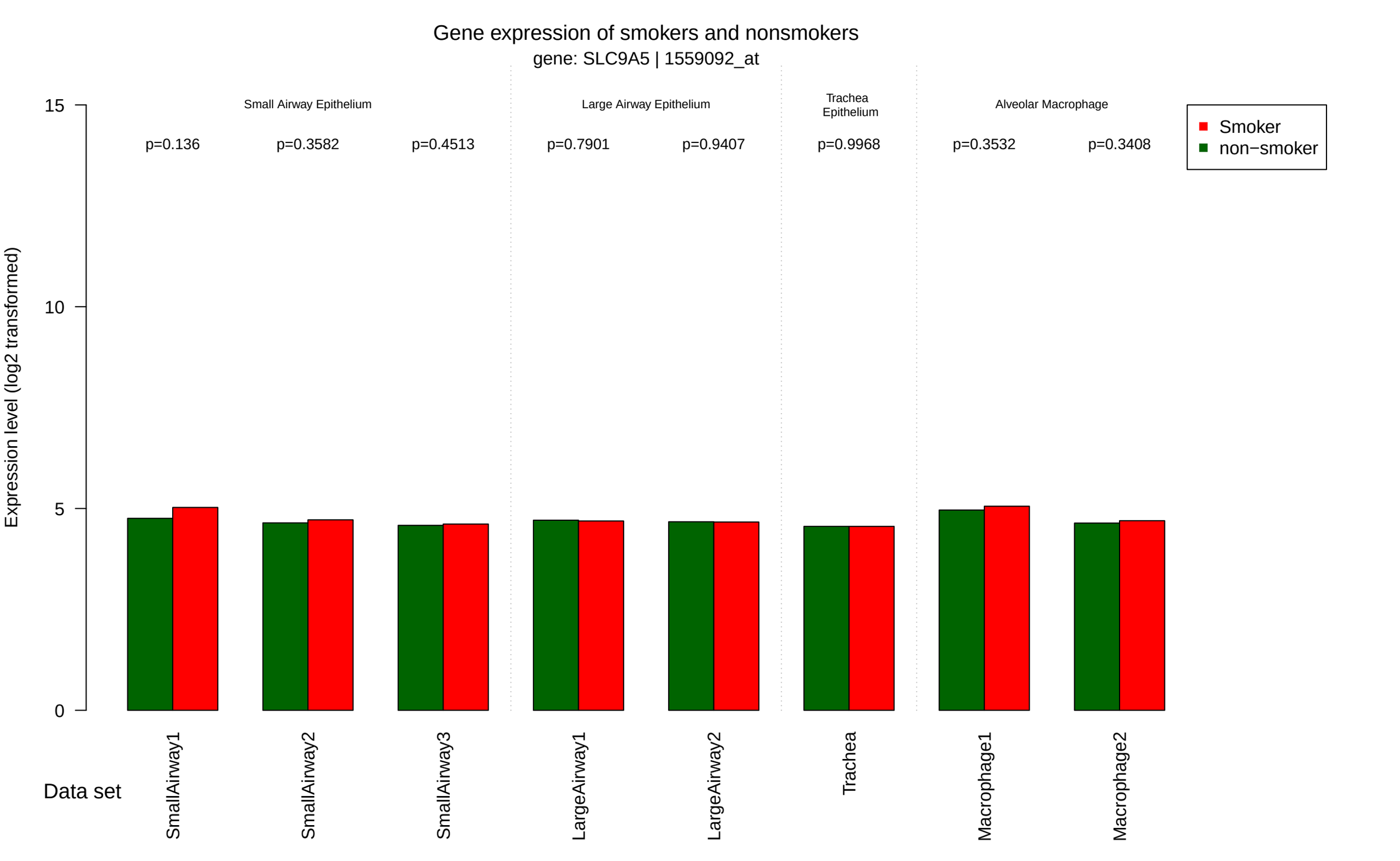
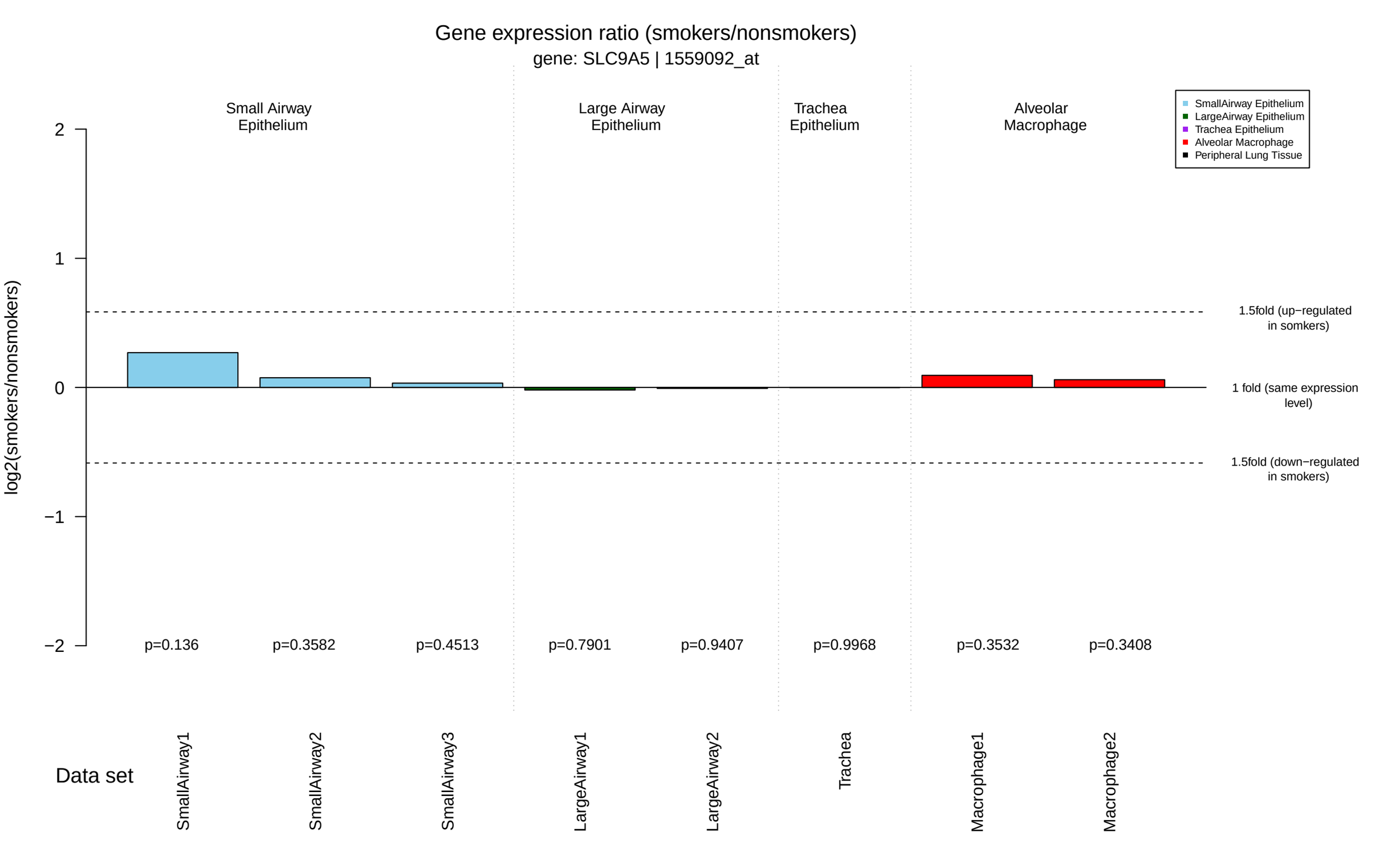
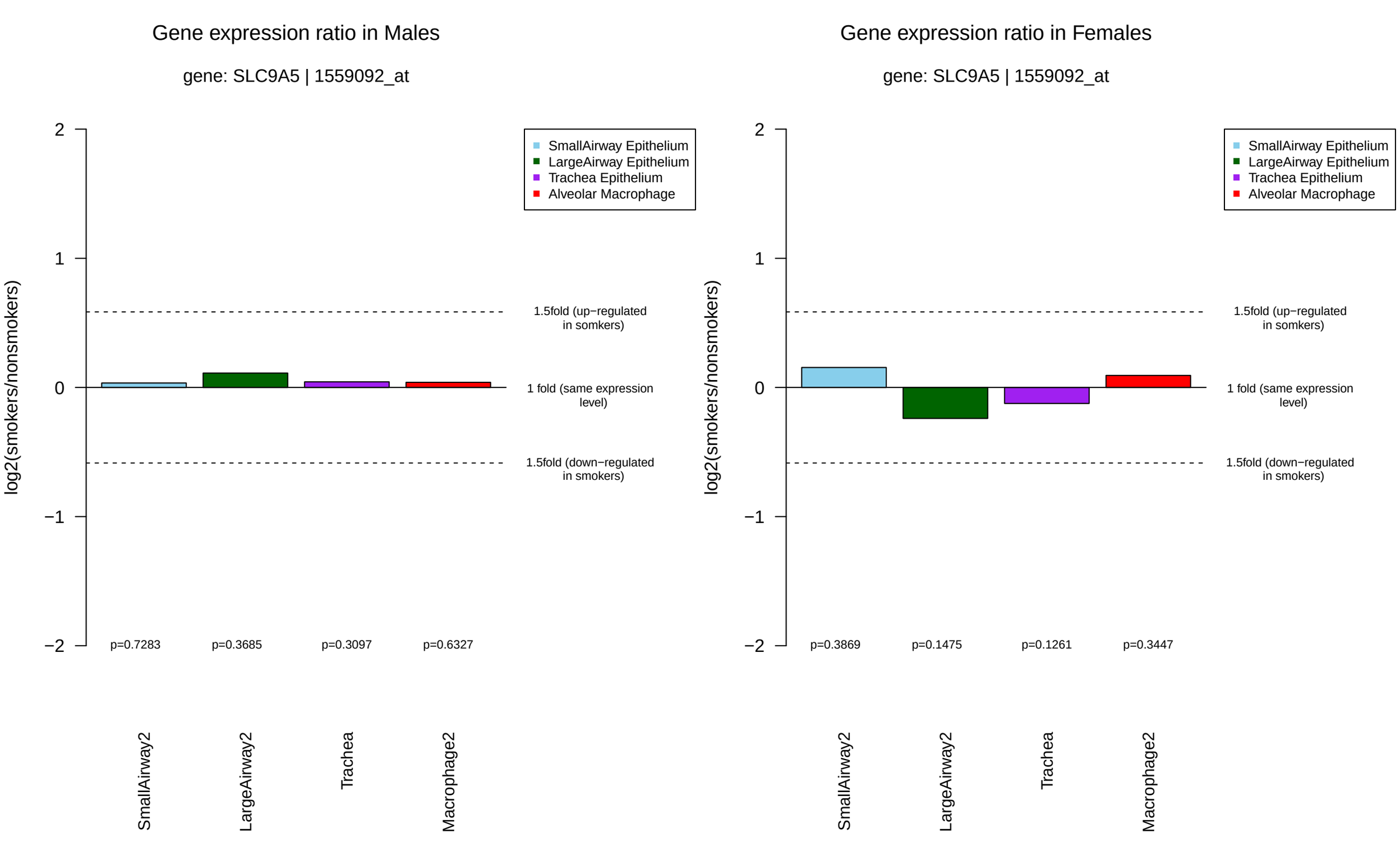
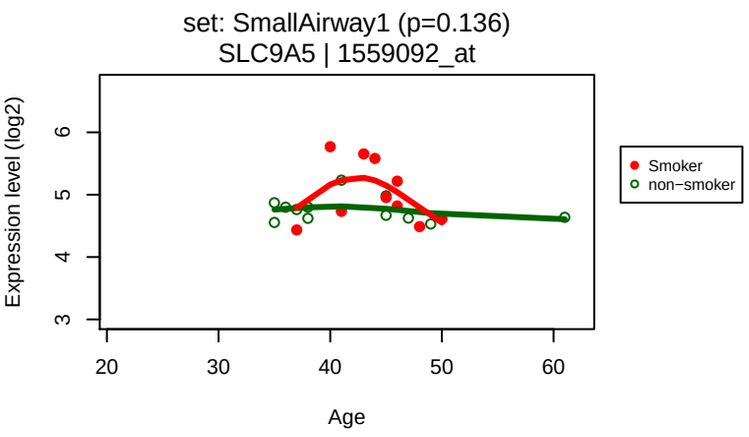
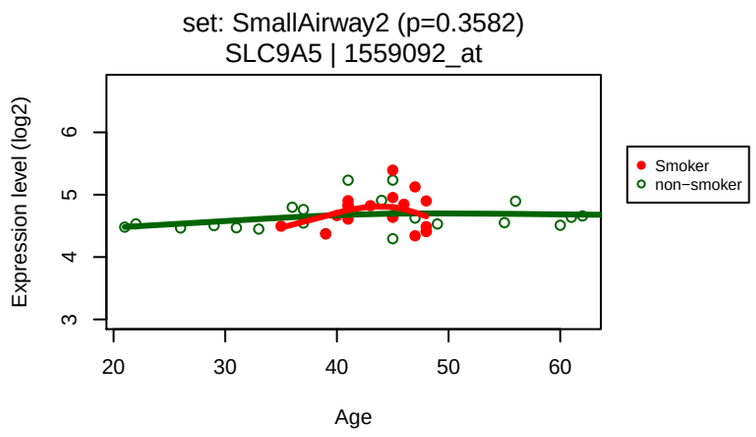
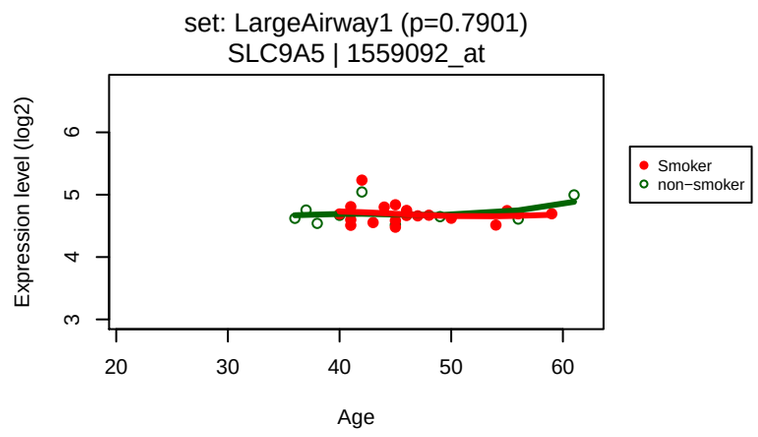

Gene expression of smokers and nonsmokers
gene: SLC9A5 | 1559092_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.136
p=0.3582
p=0.4513
p=0.7901
p=0.9407
p=0.9968
p=0.3532
p=0.3408
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

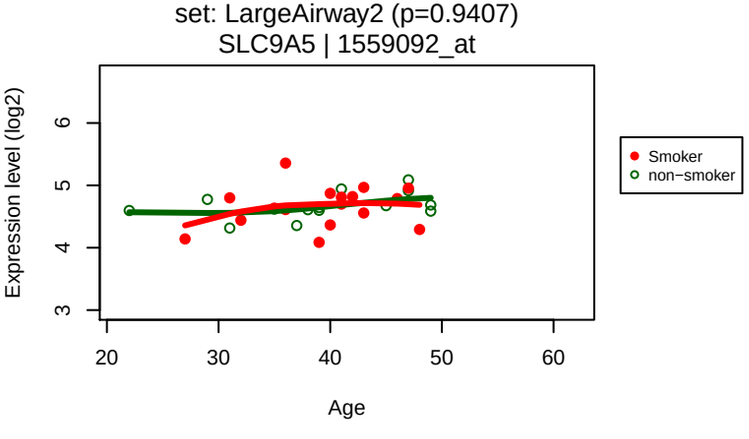

Gene expression ratio (smokers/nonsmokers)
gene: SLC9A5 | 1559092_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.136
p=0.3582
p=0.4513
p=0.7901
p=0.9407
p=0.9968
p=0.3532
p=0.3408
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

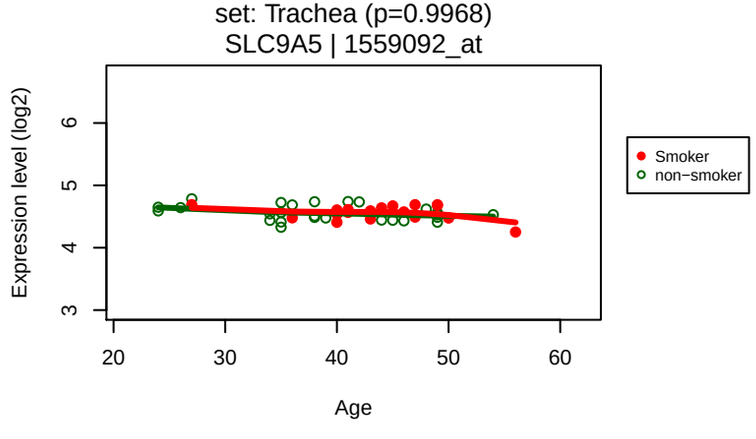

Gene expression ratio in Males
gene: SLC9A5 | 1559092_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7283
p=0.3685
p=0.3097
p=0.6327
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SLC9A5 | 1559092_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3869
p=0.1475
p=0.1261
p=0.3447
SmallAirway2
LargeAirway2
Trachea
Macrophage2

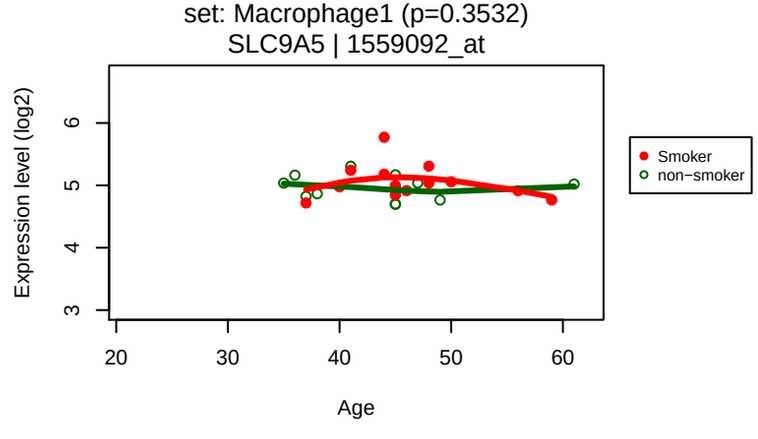

set: SmallAirway1 (p=0.136)
SLC9A5 | 1559092_at
Expression level (log2)
Age
Smoker
non−smoker

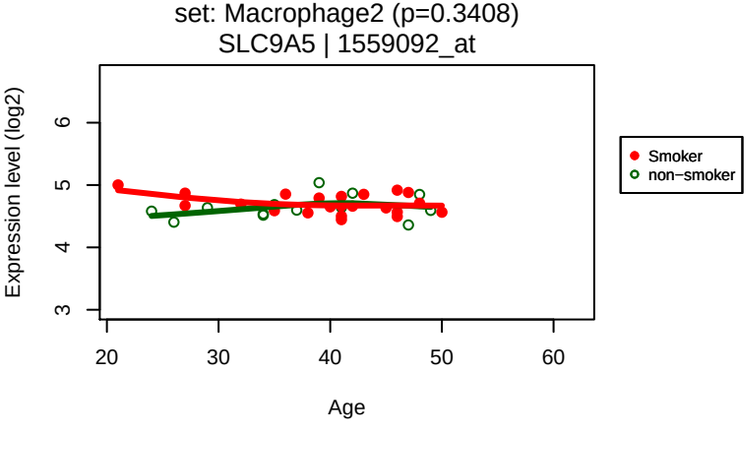

set: SmallAirway2 (p=0.3582)
SLC9A5 | 1559092_at
Expression level (log2)
Age
Smoker
non−smoker

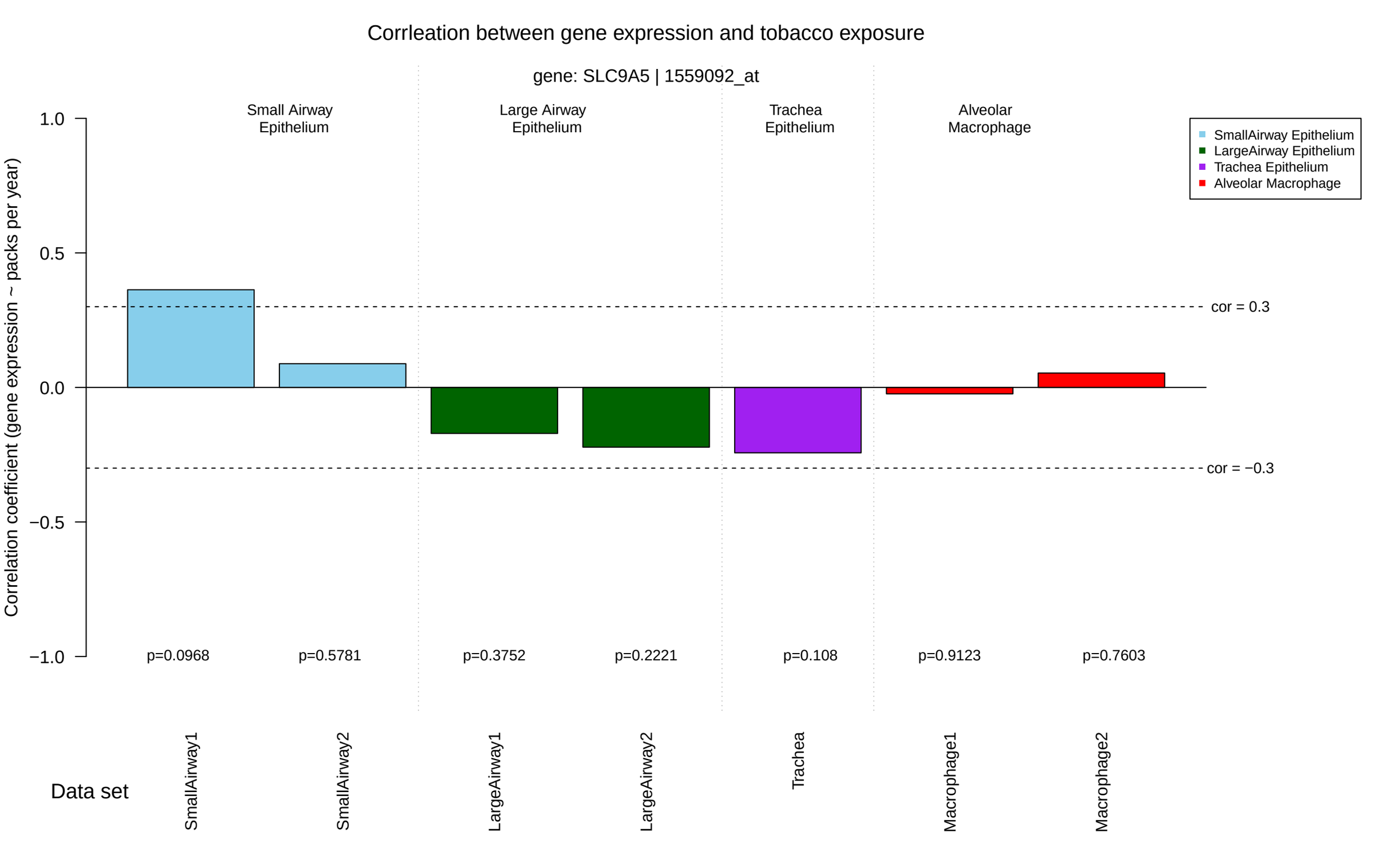

set: LargeAirway1 (p=0.7901)
SLC9A5 | 1559092_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9407)
SLC9A5 | 1559092_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9968)
SLC9A5 | 1559092_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3532)
SLC9A5 | 1559092_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3408)
SLC9A5 | 1559092_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SLC9A5 | 1559092_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0968
p=0.5781
p=0.3752
p=0.2221
p=0.108
p=0.9123
p=0.7603
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2